AIORI 3 HACKATHON • LEVEL 2 SUBMISSION

Schema-Validated YANG-Push Telemetry Pipeline

PS-034 | Cloud Computing & IoT

TEAM BEYOND LIMITS

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Abstract & problem statement

Why telemetry validation matters

THE CHALLENGE

Industrial and networked systems produce telemetry continuously. Incorrect types, malformed values, missing fields or out-of-range measurements can undermine monitoring and troubleshooting.

OUR RESPONSE

A software prototype validates simulated motor telemetry against a YANG model, applies explicitly configured safe corrections to recoverable formatting errors, revalidates the result, and records invalid notifications.

Technical background

Core concepts behind the solution

TELEMETRY

Operational measurements such as temperature, current, RPM and vibration provide visibility into equipment behaviour.

YANG

A data-modelling language used to define structured data, including expected fields, types and constraints.

YANG-PUSH

A subscription mechanism for delivering selected datastore updates. Full live YANG-Push integration is not claimed in this prototype.

Standards & technical references

Standards inform the design; implementation scope is stated separately

RFC 7950 — YANG 1.1

Defines the YANG data-modelling language.

RFC 8639 — Subscriptions

Defines a framework for subscribing to YANG notifications.

RFC 8641 — YANG-Push

Describes subscription-based delivery of datastore updates.

RFC 8259 — JSON

Defines the JSON format used for telemetry payloads in this prototype.

Proposed solution & processing workflow

Validate first • correct safely • revalidate • route outcome

01

RECEIVE

Simulated motor JSON

02

VALIDATE

Yangson + YANG model

03

CLASSIFY

Valid / recoverable / invalid

04

REVALIDATE

After configured correction

05

RECORD & DISPLAY

Accepted / rejected logs + dashboard

Correction example: “1450rpm” → 1450 (only under a predefined safe rule). “hello” as temperature → rejected.

YANG model & telemetry fields

Current industrial motor data model

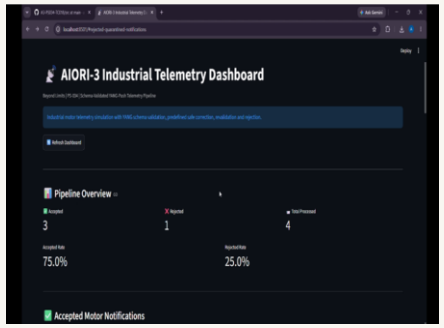
device	**string**	Motor identifier
temperature_c	**decimal64**	−50 to 200 °C
current_a	**decimal64**	0 to 1000 A
rpm	**uint32**	Non-negative integer
vibration_mm_s	**decimal64**	0 to 100 mm/s

The current validator uses the telemetry-demo.yang model and Yangson for schema-based validation.

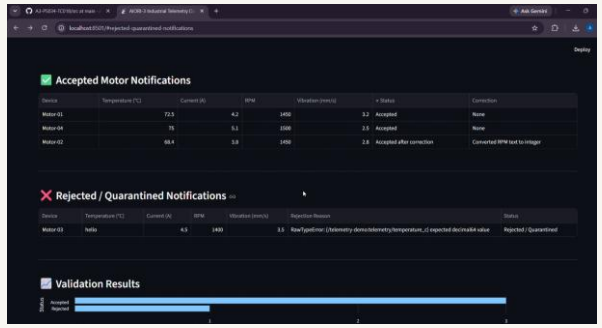
Solution architecture

Current implemented software prototype

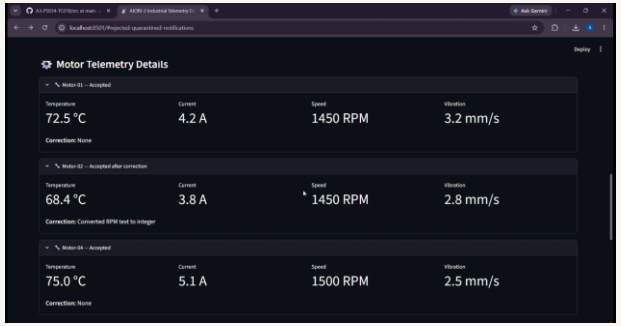
PIPELINE OVERVIEW



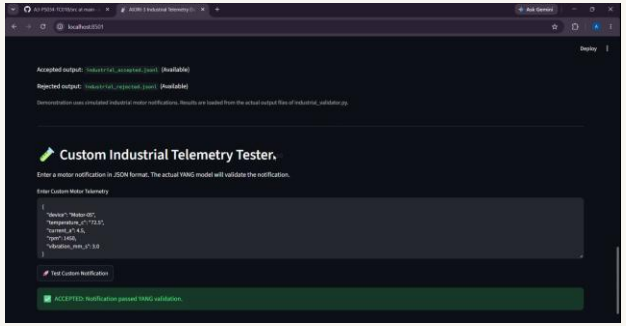
ACCEPTED & REJECTED NOTIFICATIONS



MOTOR TELEMETRY DETAILS



CUSTOM TELEMETRY TESTER



Error handling: safe correction and rejection

Only predefined, recoverable transformations are applied

RECOVERABLE FORMATTING ERROR

Input RPM: “1450rpm”

Configured rule removes the known suffix and converts the value to integer 1450.

The corrected notification is validated again before acceptance.

NON-RECOVERABLE TYPE ERROR

Input temperature: “hello”

The YANG model expects a decimal64 value.

No safe correction rule applies, so the notification is rejected and its validation error is recorded.

Monitoring dashboard

Streamlit interface for reviewing validation outcomes

OVERVIEW

Total, accepted and rejected notification counts, plus acceptance/rejection rates.

MOTOR DETAILS

Motor-wise temperature, current, RPM and vibration readings with validation status.

CUSTOM TESTER

Users can submit JSON telemetry and observe actual YANG validation results for valid and invalid inputs.

Demonstration results

Observed results from the current four-notification sample

4	3	1	75%
TOTAL	ACCEPTED	REJECTED	ACCEPTANCE RATE

3 accepted includes the notification accepted after safe correction. 1 invalid notification was rejected.

These are sample demonstration counts, not a performance benchmark or live industrial measurements.

Solution optimization

What the prototype adds to basic telemetry display

Validation	Checks schema fields, types and configured ranges
Error handling	Uses predefined safe correction; rejects non-recoverable invalid values
Traceability	Stores accepted and rejected records separately
Visibility	Provides dashboard metrics, tables, chart and custom test interface

Implementation & deployment

Current software components

PYTHON

Validation and processing logic

YANGSON

YANG schema loading and validation

STREAMLIT

Interactive dashboard

DOCKER

Containerised execution

Dashboard entry point: Dashboard.py • Validator: industrial_validator.py • Model: telemetry-demo.yang

Test cases demonstrated

Functional checks performed with simulated motor notifications

TEST 01 — VALID

A correctly typed notification satisfies the YANG model and is accepted.

TEST 02 — CORRECTION

RPM text “1450rpm” is transformed by the configured safe rule, revalidated and accepted.

TEST 03 — REJECTION

Temperature text “hello” fails decimal64 validation and is rejected with an error record.

Current scope & next development stages

Clear separation between implemented functionality and future integration

IMPLEMENTED IN THIS PROTOTYPE

- Simulated industrial motor telemetry
- Yangson-based YANG schema validation
- Predefined safe correction + revalidation
- Accepted/rejected JSONL output
- Streamlit dashboard and custom tester
- Docker-based execution

FUTURE DEVELOPMENT

- Live YANG-Push publisher or faithful emulator integration
- External message-broker integration
- End-to-end delivery and recovery tests
- Message-loss, duplicate and latency measurements
- Live industrial hardware integration

References

Technical standards and software documentation

[1] RFC 7950 — The YANG 1.1 Data Modeling Language (2016).

[2] RFC 8639 — Subscription to YANG Notifications (2019).

[3] RFC 8641 — Subscription to YANG Notifications for Datastore Updates (2019).

[4] RFC 8259 — The JavaScript Object Notation (JSON) Data Interchange Format (2017).

[5] Yangson documentation — YANG data model and instance validation.

[6] Streamlit documentation — https://docs.streamlit.io/

[7] Docker documentation — https://docs.docker.com/

Standards provide technical context; this prototype does not claim full RFC or YANG-Push protocol conformance.

THANK YOU

Schema-Validated YANG-Push Telemetry Pipeline

Questions & Discussion